AuraMed: A Unified Multi-Pathology Diagnostic Suite using Vision Transformers and Explainable AI (XAI)

1. Introduction & Problem Statement

The modern healthcare landscape faces a critical shortage of specialized diagnostic experts, particularly in rural and underserved regions. Conditions like **Breast Cancer**, **Cervical Cancer**, and **PCOS** (Polycystic Ovary Syndrome) are major health challenges for women globally, yet early screening remains inaccessible for many. Furthermore, existing AI diagnostic tools often function as "black boxes," providing a result without explaining the "why," which leads to skepticism among clinical professionals and a lack of trust in automated systems.

2. Proposed Solution

AuraMed is a comprehensive, multi-module diagnostic framework that leverages advanced deep learning to automate the detection of these three pathologies. Unlike traditional systems that focus on a single disease, AuraMed is a unified suite that provides a modular approach to medical imaging. The core innovation lies in its transition from standard Convolutional Neural Networks (CNNs) to **Vision Transformers (ViTs)**, which utilize self-attention mechanisms to identify complex, non-local pathological patterns that traditional models might miss.

3. Technical Architecture & Methodology

- **Vision Transformer (ViT-B/16) Backbone:** The model treats medical images as sequences of patches, allowing it to capture global dependencies and intricate tissue textures across the entire image.
- **The Pathological Trio:**
    - **PCOS Module:** Trained on over 3,800 ultrasound images, achieving a high-precision accuracy of **95%**.
    - **Cervical Cancer Module:** Specialized in detecting cervical dysplasia with a documented accuracy of **93.13%**.
    - **Breast Cancer Module:** A modular component designed for high-sensitivity screening of malignant vs. benign tissue structures.
- **Explainable AI (Grad-CAM):** To ensure clinical trust, we have integrated **Gradient-weighted Class Activation Mapping**. This generates visual heatmaps over the original scans, highlighting exactly where the model detected anomalies, allowing for human-in-the-loop verification.

4. Technical Implementation Abstract

The technical architecture of **AuraMed** is built upon a high-performance deep learning pipeline designed for clinical reliability and cloud scalability. At its core, the system utilizes the **Vision Transformer (ViT-B/16)** architecture, which employs a self-attention mechanism to analyze medical images as a series of patches, capturing global spatial dependencies that traditional Convolutional Neural Networks (CNNs) often overlook. This is supplemented by a **ResNet50** backbone for specialized feature extraction within the PCOS module. The entire modeling suite is developed using **PyTorch and TensorFlow**, leveraging pre-trained weights to ensure high sensitivity even with the inherent complexities of medical datasets.

To bridge the gap between AI and clinical trust, the system integrates **Explainable AI (XAI)** through **Grad-CAM**. This implementation uses **OpenCV and Matplotlib** to generate gradient-weighted class activation maps, providing real-time visual justification by highlighting pathological regions. The software layer is orchestrated using **Python**, with **FastAPI** serving as the backend gateway to manage the modular "trio" diagnostic engines.

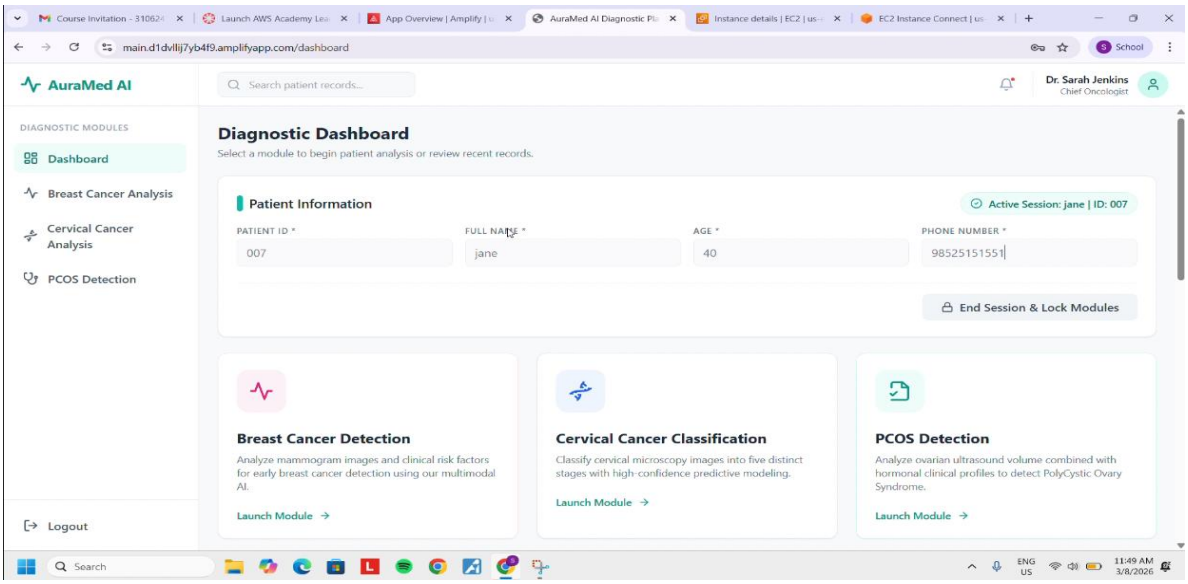
5. Cloud Scalability & Deployment

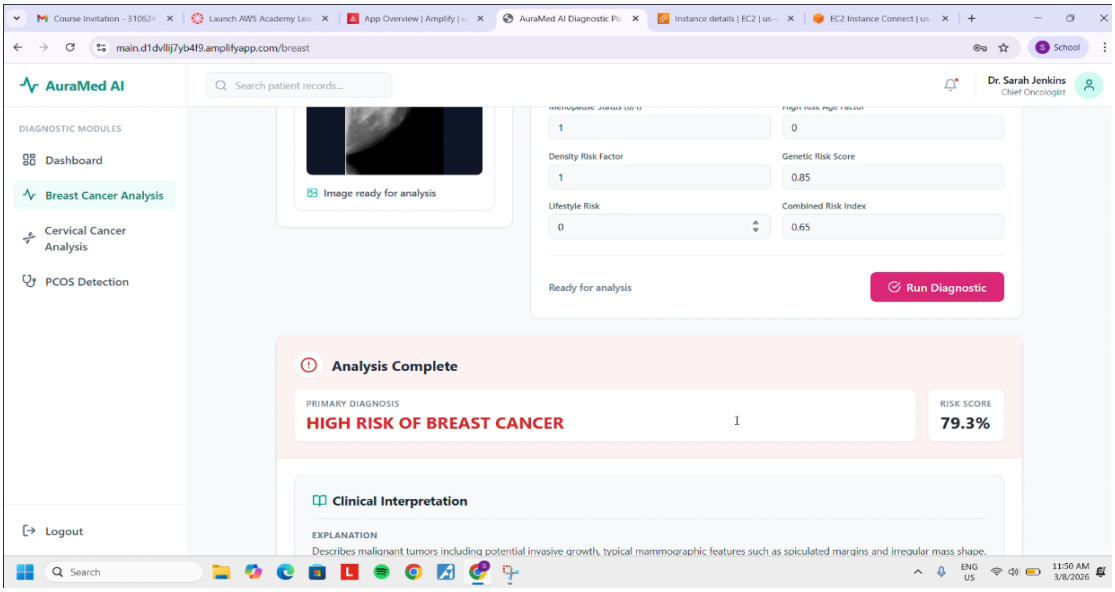
Designed with the Dept. of Cloud Computing standards in mind, AuraMed is built for seamless deployment. The modular architecture allows it to be hosted as a **Microservices-based API**. Using cloud-native features, the heavy inference of the Vision Transformer can be offloaded to GPU-accelerated instances, ensuring that a doctor in a remote location can upload an image and receive a detailed, "explained" report in seconds.

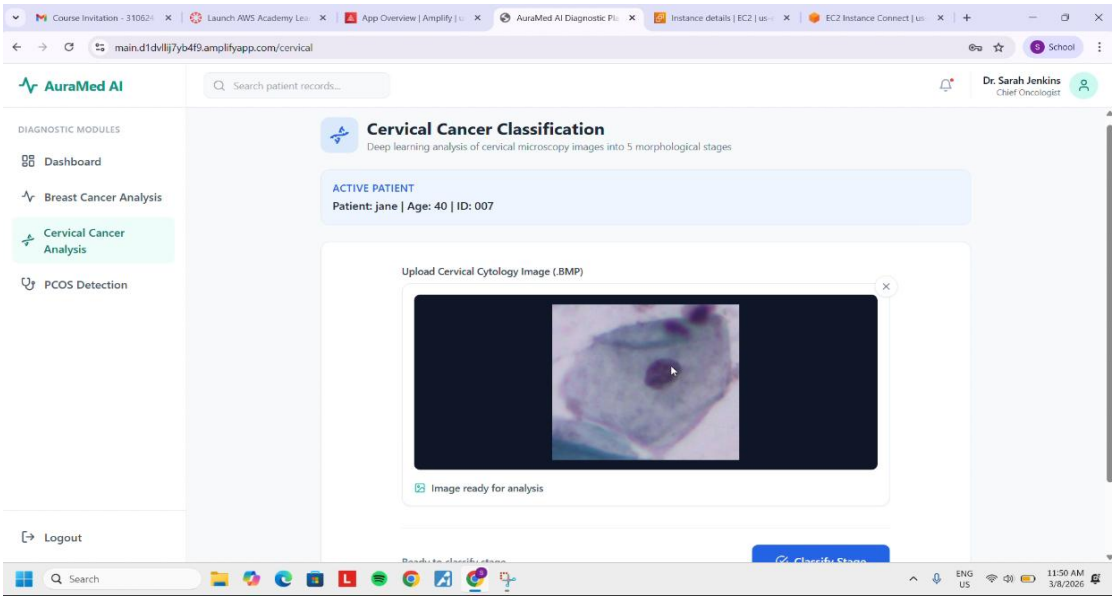
6. Impact & Sustainability

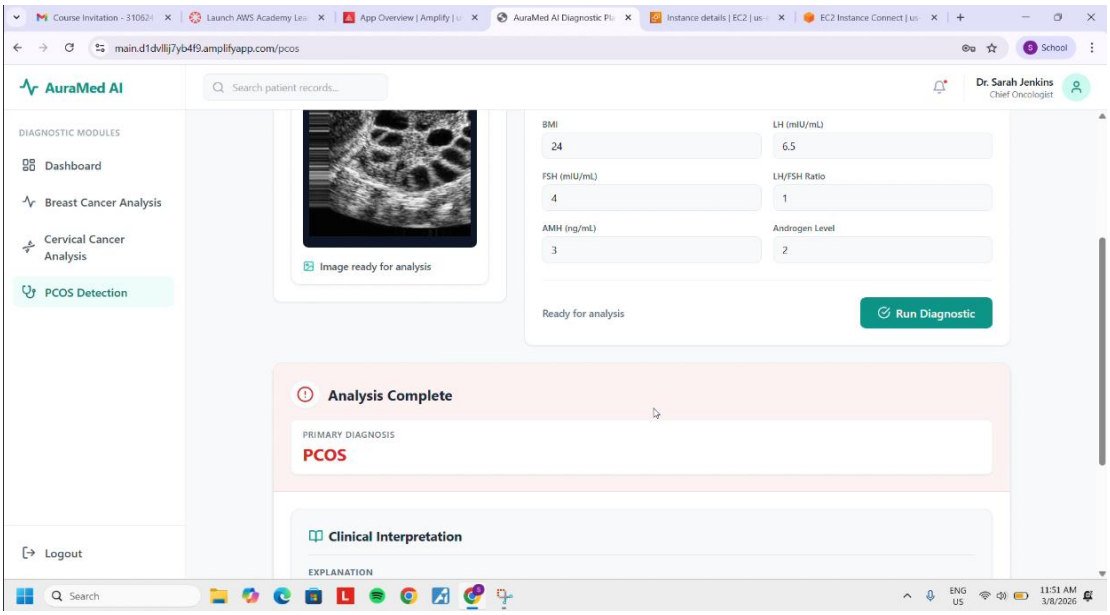
Aligned with the **UN Sustainable Development Goal 3 (Good Health and Well-being)**, AuraMed reduces the diagnostic burden on healthcare systems. It provides a low-cost, high-speed screening alternative that empowers primary care physicians and ensures that patients receive life-saving early interventions.

PROTOTYPE SCREENSHOTS:




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AuraMed AI
Search patient records...
Dr. Sarah Jenkins
Chief Oncologist
DIAGNOSTIC MODULES
Dashboard
Breast Cancer Analysis
Cervical Cancer Analysis
PCOS Detection
Logout
Image ready for analysis
Density Risk Factor
Genetic Risk Score
Lifestyle Risk
Combined Risk Index
Ready for analysis
Run Diagnostic
Analysis Complete
PRIMARY DIAGNOSIS
HIGH RISK OF BREAST CANCER
RISK SCORE
79.3%
Clinical Interpretation
EXPLANATION
Describes malignant tumors including potential invasive growth, typical mammographic features such as spiculated margins and irregular mass shape.


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Cervical Cancer Analysis
PCOS Detection
Logout
Cervical Cancer Classification
Deep learning analysis of cervical microscopy images into 5 morphological stages
ACTIVE PATIENT
Patient: jane | Age: 40 | ID: 007
Upload Cervical Cytology Image (.BMP)
Image ready for analysis


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LH/FSH Ratio
AMH (ng/mL)
Androgen Level
Ready for analysis
Run Diagnostic
Analysis Complete
PRIMARY DIAGNOSIS
PCOS
Clinical Interpretation
EXPLANATION